

SafeSphere

AI-Powered Disaster Mitigation & Emergency Response Platform



SMART INDIA
HACKATHON
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- **Problem Statement Title: SafeSphere — Disaster Response**

**Intelligence Platform for flood prediction, emergency planning,
and resource allocation.**

- **PS Category- Software**

- **Team Name: Resilience x**

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SafeSphere

Predict. Respond. Protect. — AI-powered disaster management

Detailed explanation of the proposed solution:

- A smart safety system combining AI-based risk prediction, emergency planning, response coordination, and public communication to reduce flood impact at every phase of a disaster lifecycle
- **Early Prediction:** Identifies high-risk flood situations before they become critical emergencies — ingests weather and dam-level data to detect risk conditions before they escalate
- **Dual Safety Interfaces:** Authority/rescue dashboard for command-level oversight; citizen portal for safe routes, shelter lookup, and real-time alerts — both powered by a single intelligence layer
- **Offline-Resilient Alerts:** Radio and loudspeaker-based broadcast warnings trigger automatically when internet or power connectivity is unavailable
- **What-If Scenario Simulation:** Lets leaders model rainfall impact and pre-position resources ahead of time

How it addresses the problem:

- Enables proactive rather than reactive disaster response by surfacing risk before it becomes critical
- Keeps citizens and authorities informed and coordinated even when digital infrastructure fails

Innovation and uniqueness of the solution:

- Single disaster intelligence hub serving both authority-level monitoring and citizen-facing guidance from one data layer
- Designed to align with NDMA, IMD, and Digital India frameworks for maximum institutional adoption

TECHNICAL APPROACH



◀▶ Technologies to be Used

Key Components

- **AI/ML Engine:** Predictive risk-scoring models for flood-prone zone identification, trained on weather and hydrological data
- **Data Integration APIs:** Weather forecasting APIs, dam/reservoir water-level feeds, geographic & infrastructure mapping data
- **Dashboard & Portal:** Web-based authority dashboard (React/Flutter) + citizen-facing mobile/web portal
- **Offline Broadcast System:** Automated radio and loudspeaker alert triggers, activated on connectivity/power loss
- **Cloud Backend:** Real-time data processing and storage for continuous risk monitoring across jurisdictions

== Methodology & Process

Implementation

- **System Architecture:** Weather + dam-level + geographic data → AI risk engine (scoring, early warning, What-If simulation) → dual-channel delivery (dashboard, portal, offline broadcast)
- **Step-by-Step Process:** Data ingestion → risk analysis & threshold-based alert generation → simultaneous delivery to authorities (dashboard) and citizens (portal), with automatic offline fallback
- **Independent Layer Design:** Each of the three layers (data, AI, response) operates independently, so a failure in one (e.g. internet outage) doesn't block the others
- **Prototype & Validation:** Initial demo includes live risk-scoring on sample weather/dam data with dashboard + portal alert simulation

FEASIBILITY AND VIABILITY

 Feasibility*Analysis of the Idea*

The solution is feasible because it's built on proven, established AI-based risk-assessment methods and existing digital data pipelines. The dual-interface system runs on a single unified intelligence layer, so authority and citizen tools share infrastructure instead of duplicating it — and the same platform can scale across multiple jurisdictions and agencies without rebuilding core components.

 Challenges*Potential Risks*

Key challenges include internet and power disruption during disasters, coordinating multiple emergency stakeholders across agencies, reaching remote and vulnerable populations with limited digital access, and maintaining accurate, up-to-date risk data as conditions change rapidly.

 Mitigation*Overcoming Obstacles*

We plan to overcome these challenges by automatically switching to radio and loudspeaker broadcast when connectivity fails, giving all agencies a unified dashboard with shared situational awareness, keeping the citizen portal and offline channels minimal-tech so remote communities can access them, and using predictive analysis to flag high-risk areas early — before data quality becomes a bottleneck.

IMPACT AND BENEFITS



Potential impact on the target audience:

SafeSphere targets four primary user groups across the full disaster lifecycle — from preparedness through active emergency response:

- **Citizens** in disaster-prone areas — safe routes, shelter, real-time alerts
- **Authorities** — local governments and disaster management agencies requiring centralized command intelligence
- **Rescue Teams** — live coordination maps and resource allocation tools
- **Vulnerable Communities** — remote populations depending on offline broadcast channels

Benefits of the solution:

- Faster authority decisions, improved coordination, smarter resource use
- Better preparedness through What-If scenario planning
- Reliable public communication and reduced connectivity dependence
- Remote community support and, ultimately, lives saved

RESEARCH AND REFERENCES



Details/Links of the reference and research work:

- Digital India: <https://digitalindia.gov.in/>
- Disaster Management Act, 2005: https://ndma.gov.in/Reference_Material/DMAAct2005
- IMD: <https://mausam.imd.gov.in/>
- NDMA: <https://ndma.gov.in/>
- CWC: <https://cwc.gov.in/>
- GLOFAS: <https://www.globalfloods.eu/>
- Google Flood Hydrology Framework: <https://github.com/google-research/flood-hydrology>